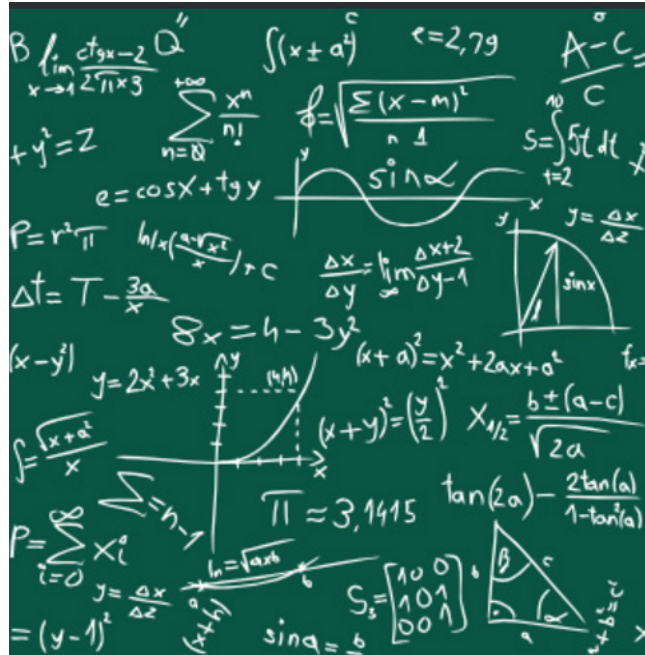


Intermediate zero theorem and bisection method



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Zeros of a function

Given a function f , we are interested in solving the equation:

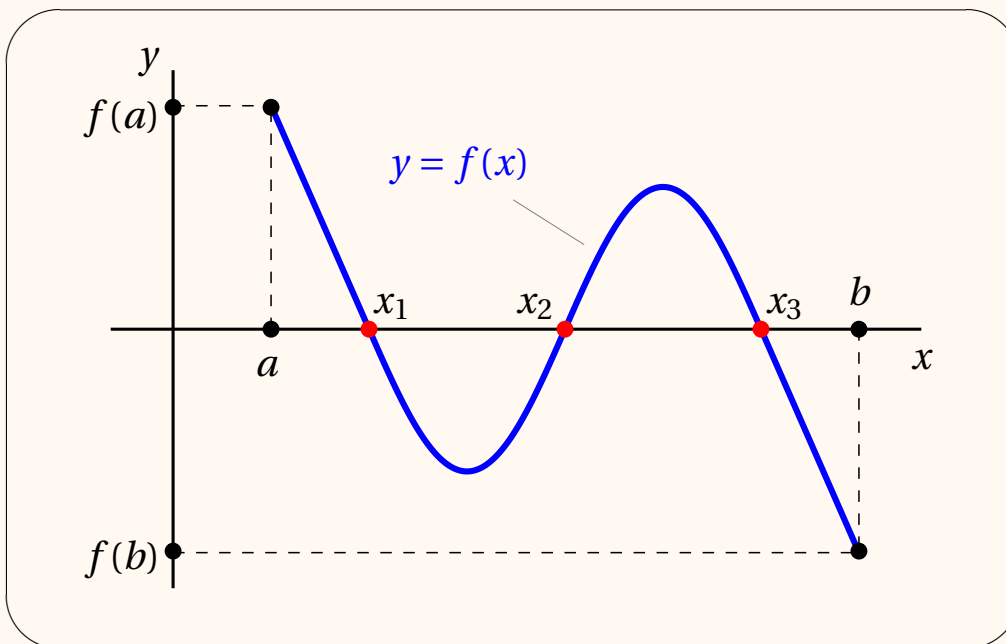
$$f(x) = 0 \quad (1)$$

that is, in finding the **zeros** of f . The zeros are the solutions of equation (1), that is, the points c of the domain of the function where $f(c) = 0$.

- When f is a polynomial of degree ≤ 4 there are formulas that give the solutions of (1). However, if f is a polynomial of degree > 4 or a more complicated function, except in particularly lucky cases, there are no formulas for the solutions of equation (1).
- Geometrically, solving equation (1) means determining the x -coordinates of the intersection points between the graph of $y = f(x)$ and the x -axis. There can be: **infinitely many solutions, a finite number of solutions, no solutions**.

Example 1: Zeros of a function

Consider the function f with the following graph:

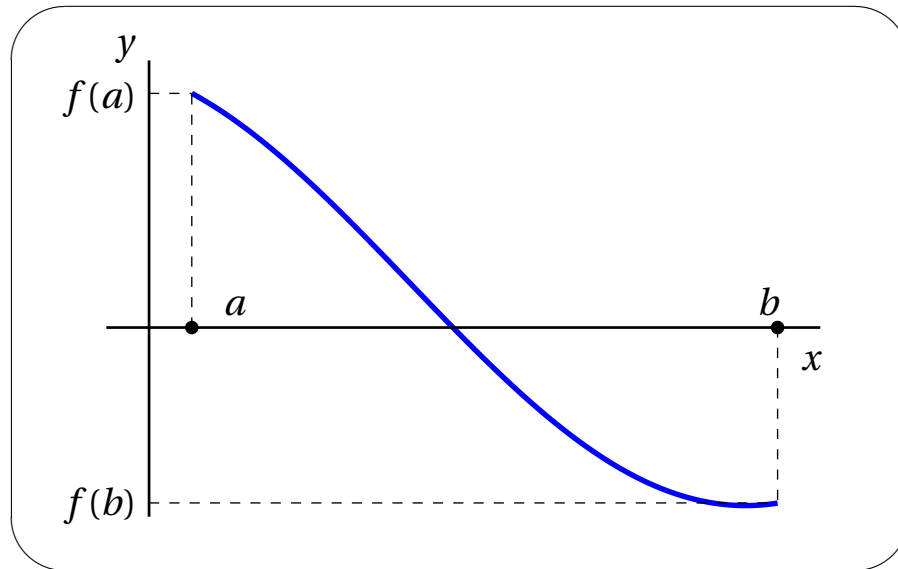


the equation $f(x) = 0$ has 3 solutions in the interval $[a, b]$, that is, the function f has 3 zeros, x_1 , x_2 and x_3 .

Theorem 1: Intermediate zero theorem (Bolzano's theorem)

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on the interval $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$. If f is also strictly monotonic then the zero c is unique.

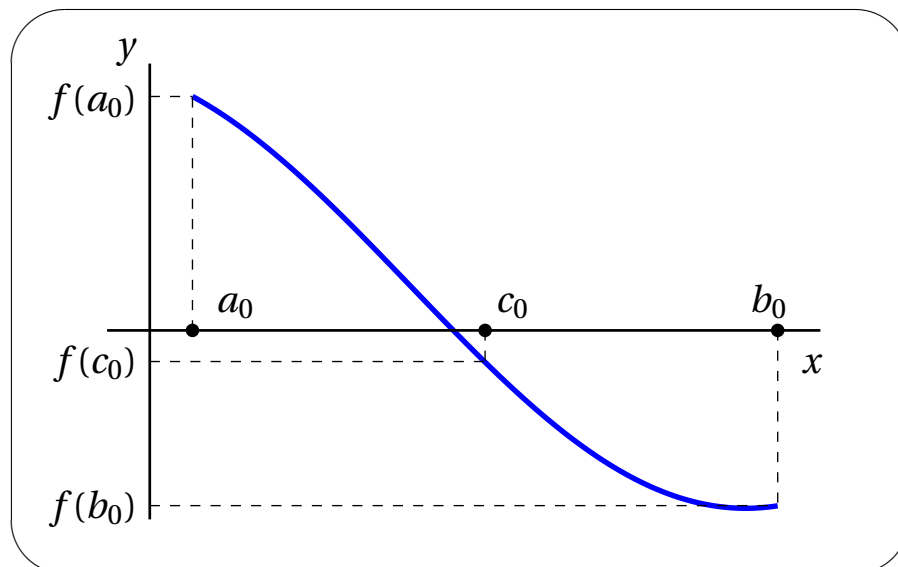
- The idea of the proof is to construct two sequences that converge to the zero of the function. To understand how to construct them, consider for example a function with $f(a) > 0$ and $f(b) < 0$:



- We set $a_0 = a$ and $b_0 = b$ and compute

$$c_0 = \frac{a_0 + b_0}{2}, \text{ the midpoint of the interval } [a_0, b_0].$$

In the previous example we have:



If $f(c_0) = 0$ we have found a zero. If $f(c_0) \neq 0$ we look at the sign of $f(a_0) \cdot f(c_0)$ and proceed as follows:

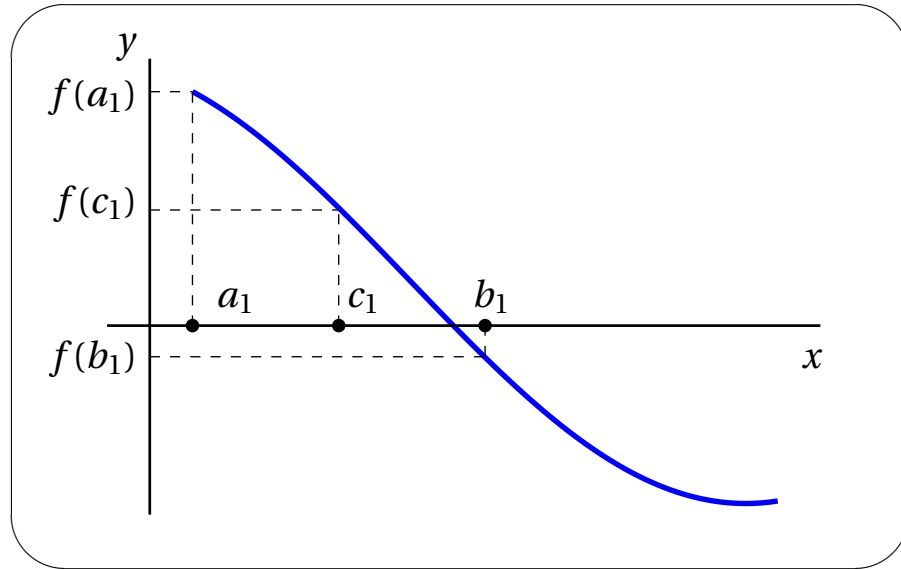
$$\left\{ \begin{array}{l} \text{if } f(a_0) \cdot f(c_0) < 0 \\ \text{if } f(a_0) \cdot f(c_0) > 0 \end{array} \right. \quad \text{we create } [a_1, b_1] \quad \left\{ \begin{array}{l} \text{with } a_1 = a_0, \quad b_1 = c_0 \\ \text{with } a_1 = c_0, \quad b_1 = b_0 \end{array} \right.$$

In the previous example we have $f(a_0) \cdot f(c_0) < 0$, hence we have $a_1 = a_0$, $b_1 = c_0$.

- With a_1 and b_1 , we compute:

$$c_1 = \frac{a_1 + b_1}{2}, \text{ the midpoint of the interval } [a_1, b_1].$$

In the previous example we have:



If $f(c_1) = 0$ we have found a zero. If $f(c_1) \neq 0$ we look at the sign of $f(a_1) \cdot f(c_1)$ and proceed as follows:

$$\left\{ \begin{array}{l} \text{if } f(a_1) \cdot f(c_1) < 0 \\ \text{if } f(a_1) \cdot f(c_1) > 0 \end{array} \right. \quad \text{we create } [a_2, b_2] \quad \left\{ \begin{array}{l} \text{with } a_2 = a_1, \quad b_2 = c_1 \\ \text{with } a_2 = c_1, \quad b_2 = b_1 \end{array} \right.$$

In the previous example we have $f(a_1) \cdot f(c_1) > 0$, hence we have $a_2 = c_1$, $b_2 = b_1$.

- We now generalize this idea in the proof of the theorem.

Proof. Consider the following two recursively defined sequences:

$$a_0 = a, \quad a_{n+1} = \begin{cases} a_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ c_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

$$b_0 = b, \quad b_{n+1} = \begin{cases} c_n & \text{if } f(a_n) \cdot f(c_n) < 0 \\ b_n & \text{if } f(a_n) \cdot f(c_n) > 0 \end{cases} \quad \forall n \in \mathbb{N}$$

where

$$c_n = \frac{a_n + b_n}{2}, \quad \forall n \in \mathbb{N}$$

The two sequences create a sequence of intervals $[a_n, b_n]$ with the following properties:

- 1) We have $a_n \leq a_{n+1}$, hence the sequence $\{a_n\}$ is increasing and, since $a_n \leq b, \forall n$, it is also bounded. Moreover, we have $b_n \geq b_{n+1}$, hence the sequence $\{b_n\}$ is decreasing and, since $b_n \geq a, \forall n$, it is also bounded.
- 2) $b_n - a_n = \frac{b-a}{2^n}$ (each interval is half as long as the previous one)
- 3) $f(a_n) \cdot f(b_n) < 0$ (because of how a_n and b_n were chosen at each step)

By point 1), we can then deduce that the sequences $\{a_n\}$ and $\{b_n\}$ have a finite limit, thanks to the monotone sequence theorem. Hence:

$$a_n \rightarrow \ell_1 \in \mathbb{R} \text{ and } b_n \rightarrow \ell_2 \in \mathbb{R} \text{ as } n \rightarrow +\infty.$$

From point 2) we deduce that:

$$b_n - a_n = \frac{b-a}{2^n} \rightarrow 0 \text{ as } n \rightarrow +\infty, \text{ and therefore } \ell_2 = \ell_1 = \ell$$

By the continuity of f , we then have that:

$$f(a_n) \cdot f(b_n) \rightarrow (f(\ell))^2 \text{ as } n \rightarrow +\infty$$

On the other hand, from point 3) and the sign-preservation theorem we deduce $(f(\ell))^2 \leq 0$. Therefore it must be $f(\ell) = 0$ and thus ℓ is the zero we are looking for, that is, $c = \ell$.

□

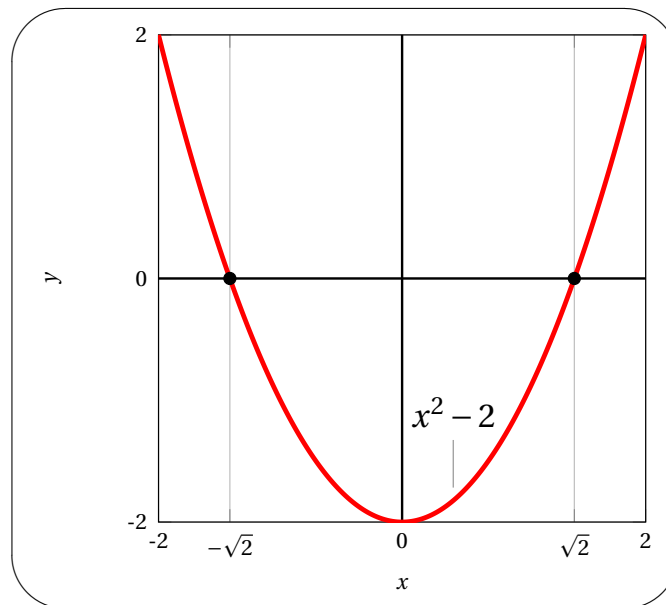
- We have proved:

$f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0 \Rightarrow$ there exists $c \in [a, b]$ such that $f(c) = 0$

hence “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ” is a sufficient condition for “there exists $c \in [a, b]$ such that $f(c) = 0$ ” and “there exists $c \in [a, b]$ such that $f(c) = 0$ ” is a necessary condition for “ $f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$ ”. The implication does not work in the other direction:

there exists $c \in [a, b]$ such that $f(c) = 0 \not\Rightarrow f : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$

It suffices, for example, to consider $f(x) = x^2 - 2$ on the interval $[-2, 2]$. We have $f(-2) = 2$ and $f(2) = 2$, hence $f(a) \cdot f(b) \not< 0$, but the function does have zeros:



Hence the theorem provides **sufficient** (but not necessary) conditions for the existence of a zero of a function.

- The proof of the theorem is constructive, that is, we have constructed two sequences that tend to a zero of the function (**constructive proof/algorithm**). This algorithm is called the **bisection method**.

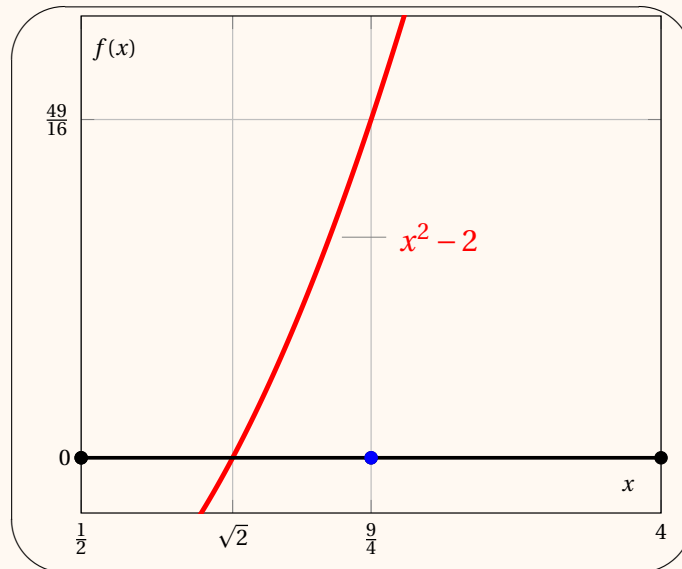
If the function f has several zeros in $[a, b]$, the procedure does not indicate which of them will be determined. The zero found depends on the interval $[a, b]$ given as input, and finding it may require infinitely many iterations. However, by stopping the procedure, as we will see, we obtain an estimate of the zero and of the error made.

2 Bisection method

Example 2: Bisection method

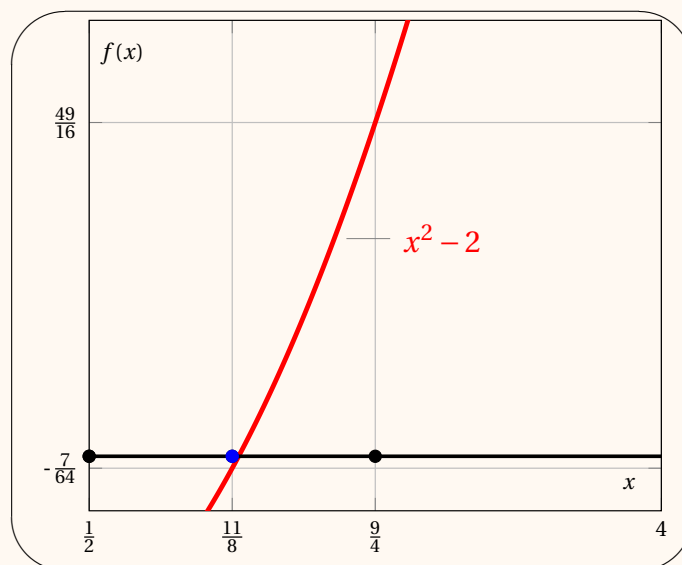
- We look for a zero of the continuous function $f(x) = x^2 - 2$ on the interval $[\frac{1}{2}, 4]$, that is, we look for the value of $\sqrt{2}$. Initialization, with $n = 0$ we have:

$$[a_0, b_0] = \left[\frac{1}{2}, 4\right], \quad f(a) = -\frac{7}{4}, \quad f(b) = 14 \Rightarrow c_0 = \frac{9}{4} \text{ and } f(c_0) = \frac{49}{16}$$



- Iteration $n = 1$, we have:

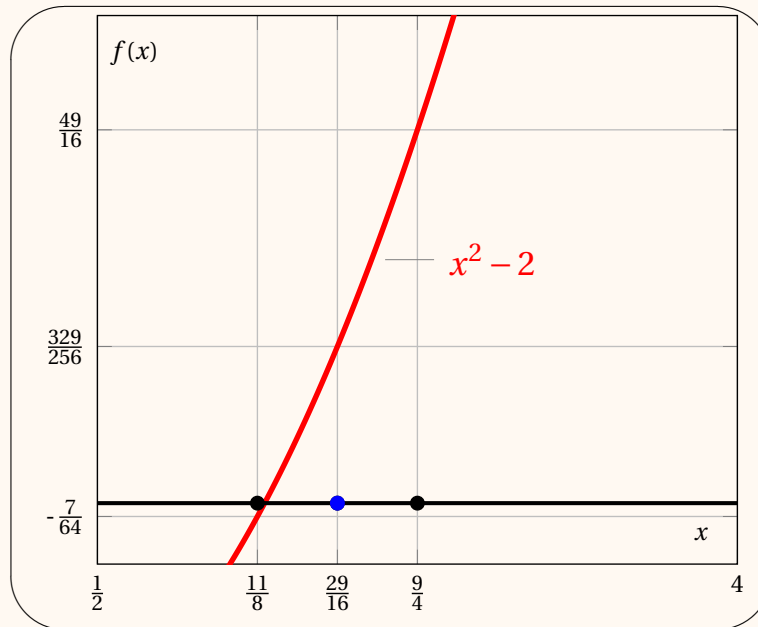
$$[a_1, b_1] = \left[\frac{1}{2}, \frac{9}{4}\right], \quad f(a_1) = -\frac{7}{4}, \quad f(b_1) = \frac{49}{16} \Rightarrow c_1 = \frac{11}{8} \text{ and } f(c_1) = -\frac{7}{64}$$



Example 3: Bisection method

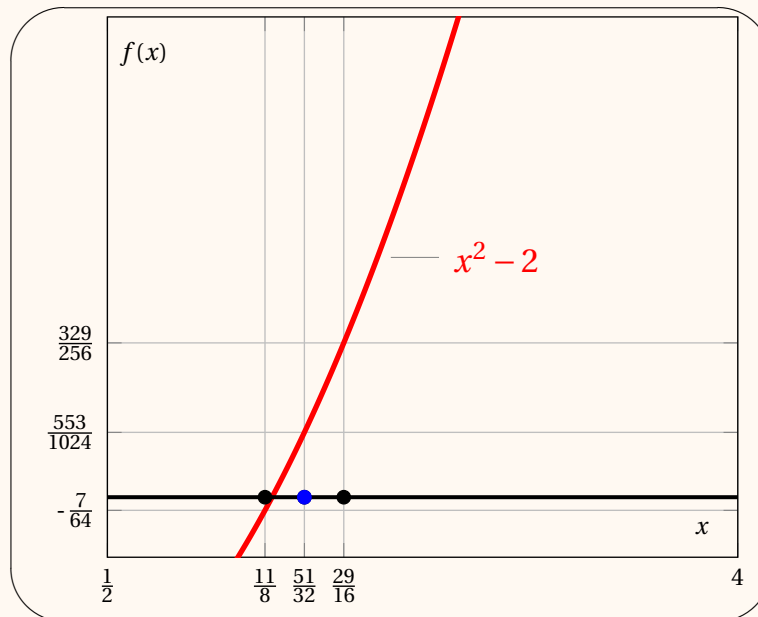
- Iteration $n = 2$, we have:

$$[a_2, b_2] = \left[\frac{11}{8}, \frac{9}{4} \right], \quad f(a_2) = -\frac{7}{64}, \quad f(b_2) = \frac{49}{16} \Rightarrow c_2 = \frac{29}{16} \text{ and } f(c_2) = \frac{329}{256}$$



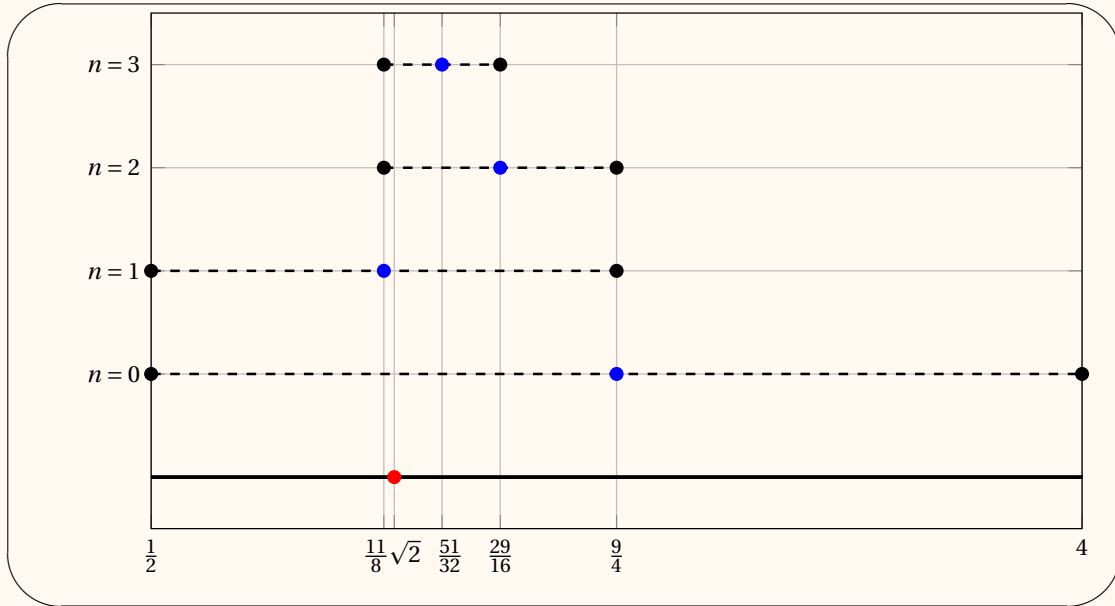
- Iteration $n = 3$, we have:

$$[a_3, b_3] = \left[\frac{11}{8}, \frac{29}{16} \right], \quad f(a_3) = -\frac{7}{64}, \quad f(b_3) = \frac{329}{256} \Rightarrow c_3 = \frac{51}{32} \text{ and } f(c_3) = \frac{553}{1024}$$



Example 4: Intervals of the bisection method

Plotting the values obtained in the previous exercise, we have the following intervals and estimates of $\sqrt{2}$:



2.1 Error estimates for the bisection method

- At iteration n , we have:

$$c \in [a_n, b_n]$$

that is, the zero lies inside the n -th interval. Hence, stopping the procedure after n steps, a_n approximates the zero from below and b_n approximates the zero from above. Therefore:

$$a_n \leq c \leq b_n$$

- The estimate of the value c at iteration n is:

$$c_n = \frac{a_n + b_n}{2}$$

that is, the midpoint of the interval $[a_n, b_n]$.

- The absolute error ε_n made at iteration n is:

$$\varepsilon_n = |c_n - c|$$

The value of c is unknown, hence we need a way to analyze the values of the errors that is independent of c .

- We can estimate the error at iteration n as follows:

$$\varepsilon_n = |c_n - c| \leq \frac{b-a}{2^n}$$

this follows from:

- the length of the interval at iteration n is $\frac{b-a}{2^n}$
- the midpoint of this interval is c_n and c lies inside the interval itself
- the distance of any point c from the center of the interval is less than or equal to the length of the interval itself

If $n \rightarrow +\infty$ then $\varepsilon_n \rightarrow 0$, that is, the error tends to zero as n tends to infinity.

- At iteration n , we can improve the error estimate as follows:

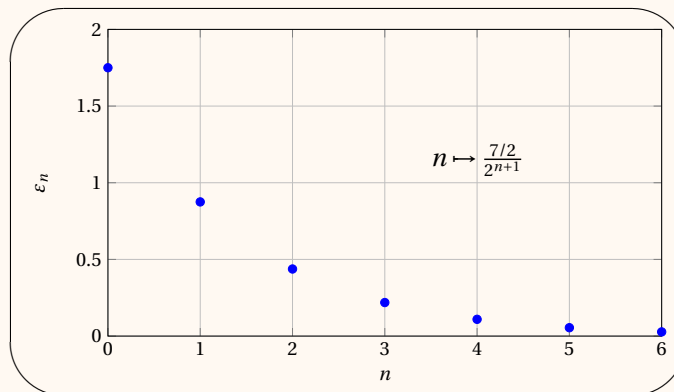
$$\varepsilon_n = |c_n - c| \leq \frac{1}{2} \frac{b-a}{2^n} = \frac{b-a}{2^{n+1}}$$

that is, half the length of the interval $[a_n, b_n]$, since $c \in (a_n, c_n)$ or $c \in (c_n, b_n)$.

Example 5: Error estimates for the bisection method ($\sqrt{2} = 1.414213562\dots$)

Let us go back to the previous exercise with $b-a = \frac{7}{2}$; in the first 6 iterations we have:

iteration	a_n	c_n	b_n	estimate of $\sqrt{2}$	ε_n
$n=0$	$\frac{1}{2}$	$\frac{9}{4}$	4	2.25	$\leq \frac{7/2}{2^1} = 1.75$
$n=1$	$\frac{1}{2}$	$\frac{11}{8}$	$\frac{9}{4}$	1.375	$\leq \frac{7/2}{2^2} = 0.875$
$n=2$	$\frac{11}{8}$	$\frac{29}{16}$	$\frac{9}{4}$	1.8125	$\leq \frac{7/2}{2^3} = 0.4375$
$n=3$	$\frac{11}{8}$	$\frac{51}{32}$	$\frac{29}{16}$	1.59375	$\leq \frac{7/2}{2^4} = 0.21875$
$n=4$	$\frac{11}{8}$	$\frac{95}{64}$	$\frac{51}{32}$	1.484375	$\leq \frac{7/2}{2^5} = 0.109375$
$n=5$	$\frac{11}{8}$	$\frac{183}{128}$	$\frac{95}{64}$	1.4296875	$\leq \frac{7/2}{2^6} = 0.0546875$



- In order to guarantee that the error made does not exceed a given tolerance δ , that is, to impose $\varepsilon_n \leq \delta$, we need to perform $n(\delta)$ iterations, where $n(\delta)$ is the smallest integer that satisfies the inequality:

$$n(\delta) > \log_2 \left(\frac{b-a}{\delta} \right) - 1$$

obtained by solving for $n(\delta)$ in the maximum error estimate:

$$\delta = \frac{b-a}{2^{(n(\delta)+1)}}$$

hence

$$2^{(n(\delta)+1)} = \frac{b-a}{\delta} \quad \text{and} \quad n(\delta) + 1 = \log_2 \left(\frac{b-a}{\delta} \right)$$

Clearly, if $\delta \rightarrow 0$ then $n(\delta) \rightarrow +\infty$, and the smaller the error tolerance, the larger the number of iterations needed to guarantee it.

- If we wanted to reduce the tolerance by one decimal digit, that is, to go from δ to $\frac{\delta}{10}$, we would have:

$$n(\delta) \approx \log_2 \left(\frac{b-a}{\delta} \right) - 1 \quad \text{and} \quad n\left(\frac{\delta}{10}\right) \approx \log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1$$

since

$$\log_2 \left(\frac{b-a}{\frac{\delta}{10}} \right) - 1 = \log_2 \left(\frac{b-a}{\delta} \cdot 10 \right) - 1 = \underbrace{\log_2 \left(\frac{b-a}{\delta} \right) - 1}_{\approx n(\delta)} + \log_2 10$$

the number of additional iterations is independent of the interval $[a, b]$, and it is equal to

$$\log_2 10 \approx 3,32$$

On average, more than three bisections are needed to improve the accuracy of the estimate by one significant digit; **convergence is therefore slow.**

Example 6: Bisection method (error estimates)

Again for the previous example:

- if we want a tolerance (maximum error) $\delta = 0.00001 = 10^{-5}$:

$$n(10^{-5}) > \log_2 \left(\frac{7/2}{10^{-5}} \right) - 1 = 17.416... \quad \text{hence} \quad n(10^{-5}) = 18$$

- if we want a tolerance (maximum error) $\delta = 0.000001 = 10^{-6}$:

$$n(10^{-6}) > \log_2 \left(\frac{7/2}{10^{-6}} \right) - 1 = 20.738... \quad \text{hence} \quad n(10^{-6}) = 21$$

We now want to understand how “fast” the error decreases by comparing the errors of two consecutive iterations. This estimate gives us information about the quality of the algorithm, that is, we want to determine the “**rate of convergence**” of the algorithm.

- Consider the maximum error, which we denote by ε_n^* ; we have:

$$\varepsilon_n^* = \frac{b-a}{2^{n+1}} \quad \text{and} \quad \varepsilon_n \leq \varepsilon_n^*$$

hence

$$\frac{\varepsilon_{n+1}^*}{\varepsilon_n^*} = \frac{1}{2} \quad \text{since} \quad \frac{\frac{b-a}{2^{n+2}}}{\frac{b-a}{2^{n+1}}} = \frac{1}{2}$$

that is, the maximum error is halved at each iteration. Consequently, we have:

$$\underbrace{\varepsilon_{n+1}^*}_{\geq \varepsilon_{n+1}} = \frac{1}{2} \underbrace{\varepsilon_n^*}_{\geq \varepsilon_n}$$

The maximum absolute error of the bisection method at each step is proportional to the maximum absolute error at the previous step. **The rate of convergence is linear.**

- For every triple c_{n+1}, c_n and c_{n-1} , with $n \geq 1$, we have the following relation:

$$c_{n+1} = \frac{c_{n-1} + c_n}{2} \tag{2}$$

that is, the estimate c_{n+1} is the midpoint of the interval with endpoints c_{n-1} and c_n . We now write:

$$c_{n+1} = c + \varepsilon_{n+1}, \quad c_n = c + \varepsilon_n, \quad c_{n-1} = c + \varepsilon_{n-1}$$

now, substituting into equation (2), we obtain:

$$\begin{aligned} c + \varepsilon_{n+1} &= \frac{c + \varepsilon_n + c + \varepsilon_{n-1}}{2} = \frac{2c + \varepsilon_n + \varepsilon_{n-1}}{2} \\ &= c + \frac{\varepsilon_n + \varepsilon_{n-1}}{2} \end{aligned}$$

hence

$$\varepsilon_{n+1} = \frac{\varepsilon_n + \varepsilon_{n-1}}{2} = \frac{1}{2} \varepsilon_n \left(\frac{\varepsilon_{n-1}}{\varepsilon_n} + 1 \right)$$

The term $\varepsilon_{n-1}/\varepsilon_n$, that is, the ratio between the errors of two consecutive iterations, can be an arbitrarily small or large number (indeed, the error at each iteration can increase or decrease).